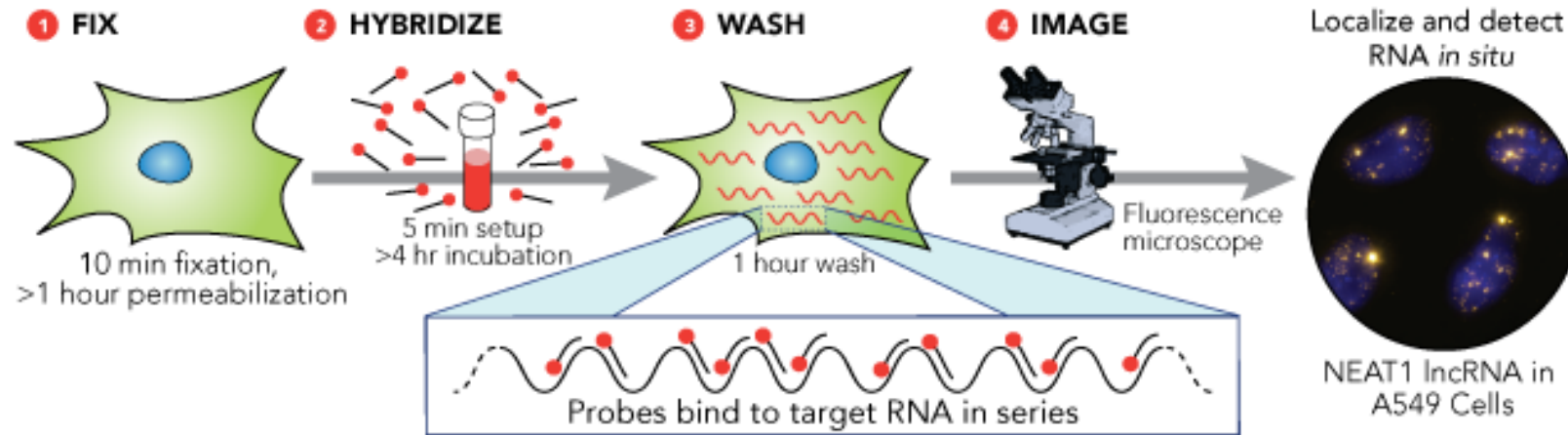


Data pipelines for FISH imaging

Jacques Bourg

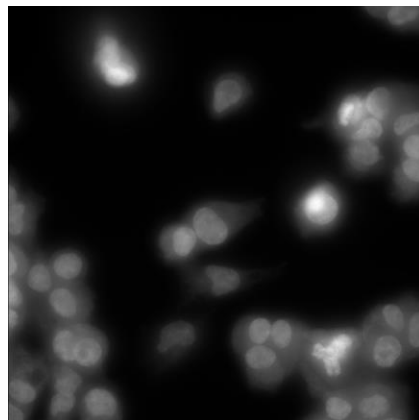
FISH: fluorescent in situ hybridization



Scientific context:

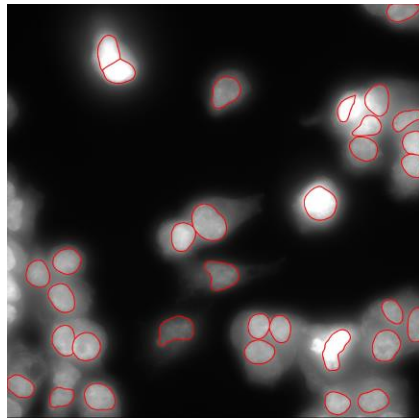
- HOX gene expression under different growth factors.
- Stress responses under LNP.

Max projection

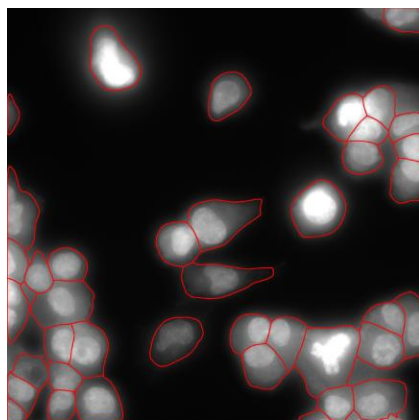


Segmentation

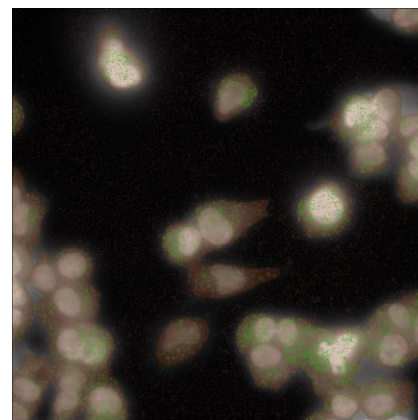
Nuclei



Cells

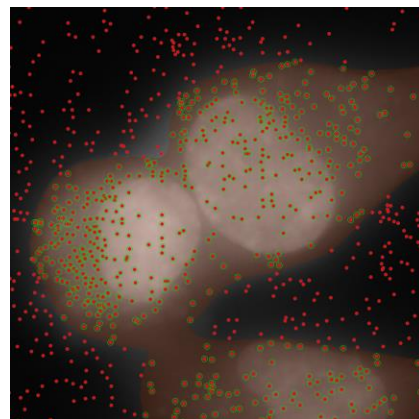


Binary masks



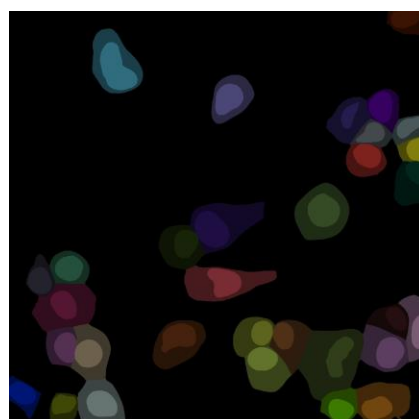
RNA detection

Spot detection and cleaning

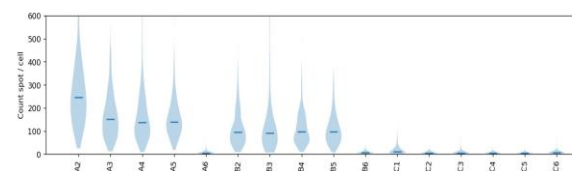


Postprocessing

RNA: cell assignment



Statistics



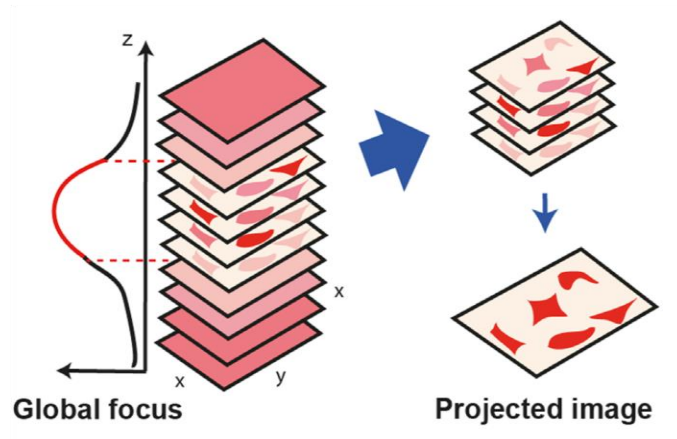
Spatial RNA descriptors

Intensity descriptors

Max intensity projection

Max projection

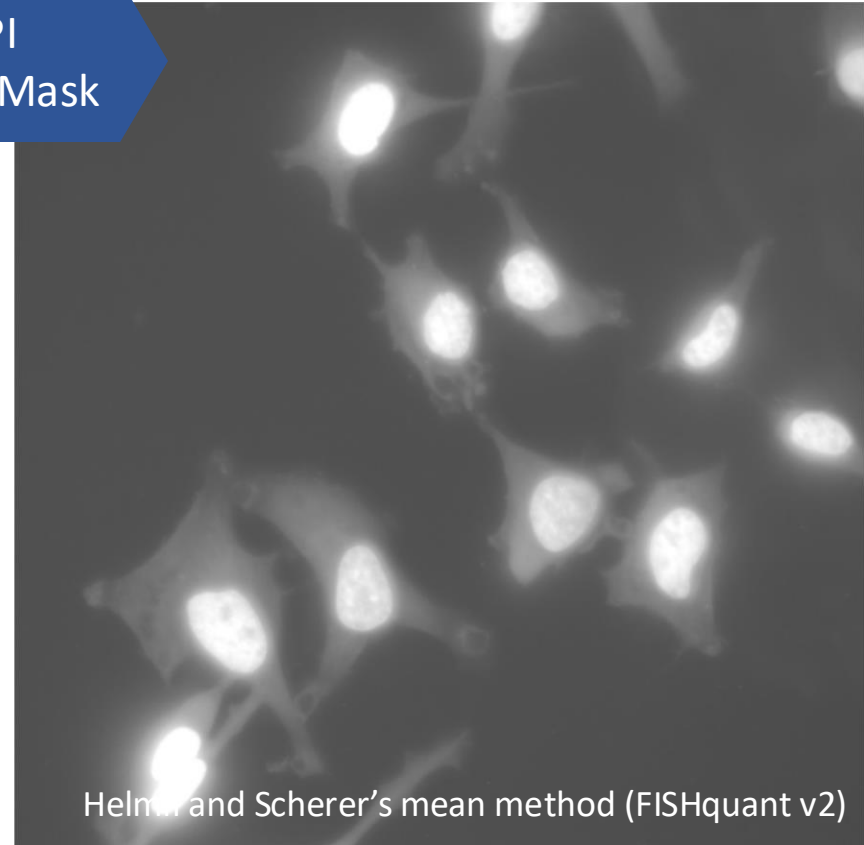
With focus measurement



Tsanov et al, 2016, NAR

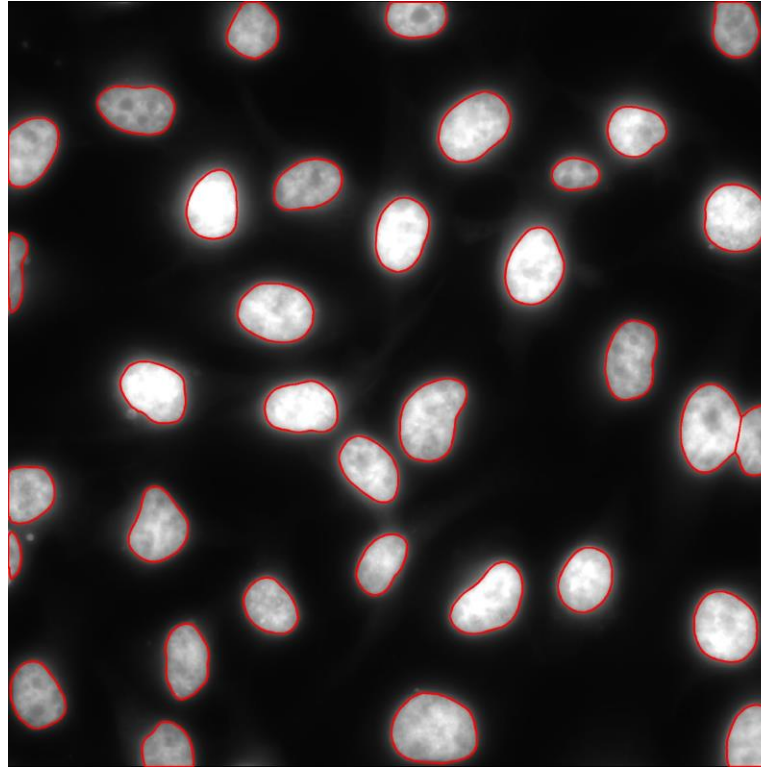
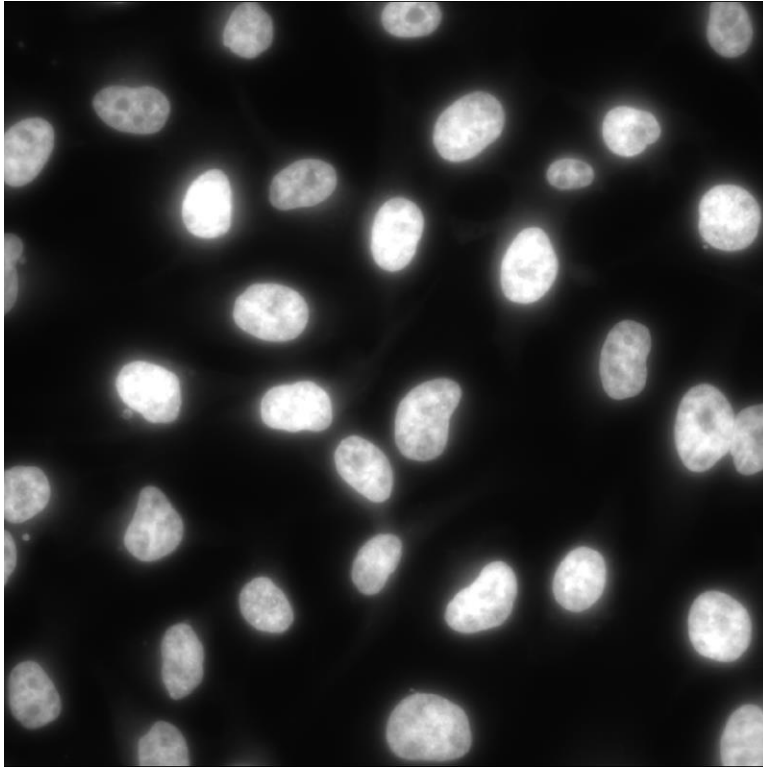
Imbert et al, 2022, RNA

DAPI
CellMask



Helmer and Scherer's mean method (FISHquant v2)

Segmentation



Human in the loop approach:

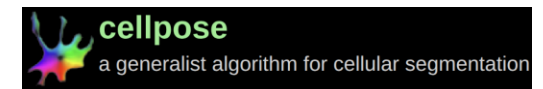
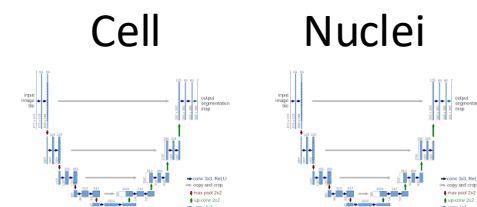
Starting from a specialized model (nuclei or cyto2):

- 1) Apply it to a new image.
- 2) Correct manually the segmentation errors.
- 3) Retrain (using current and precedent segmentations).

Iterate 1-3 until the segmentation quality is good for a newly seen image.

We obtain a custom model for our cells. No need of training/test database.

- Segment nuclei from combined 'blue' image
- Cellpose 3 with minimal retraining
- Critical parameter for training/ inference: diameter.
- Train one model for the nuclei and one model for the cell bodies.

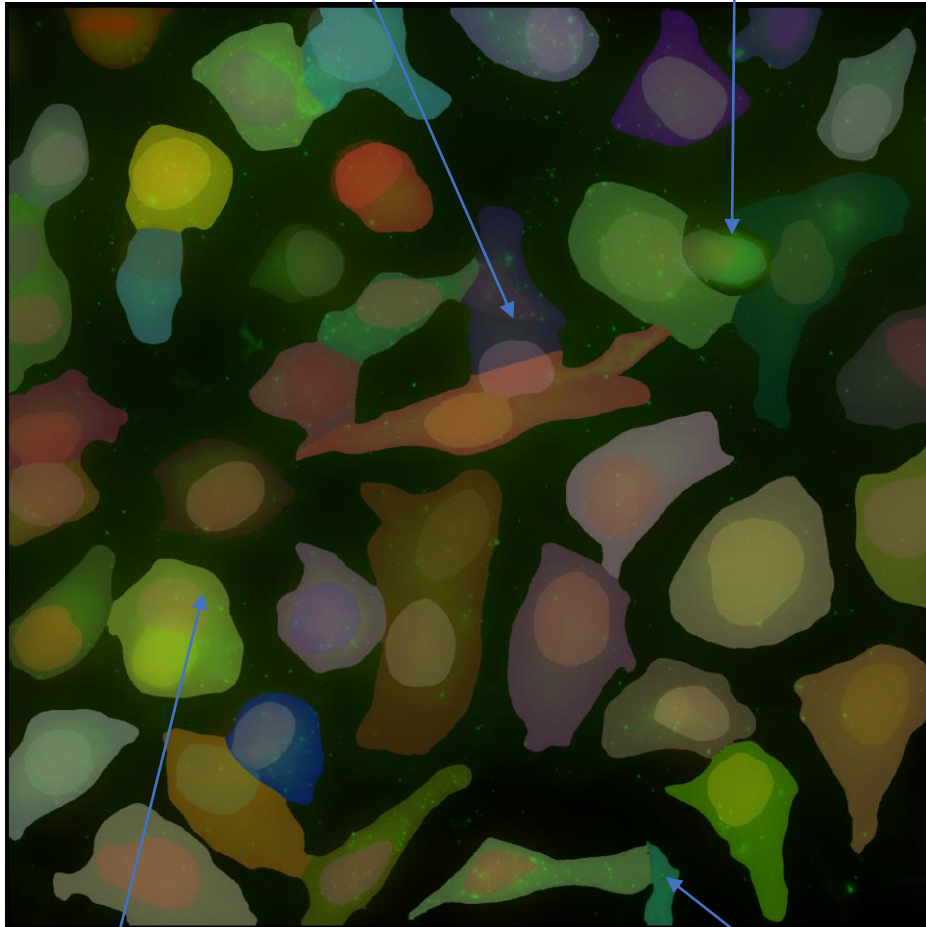


Segmentation postprocessing (I/II)

Nuclei mask extends
beyond cell mask

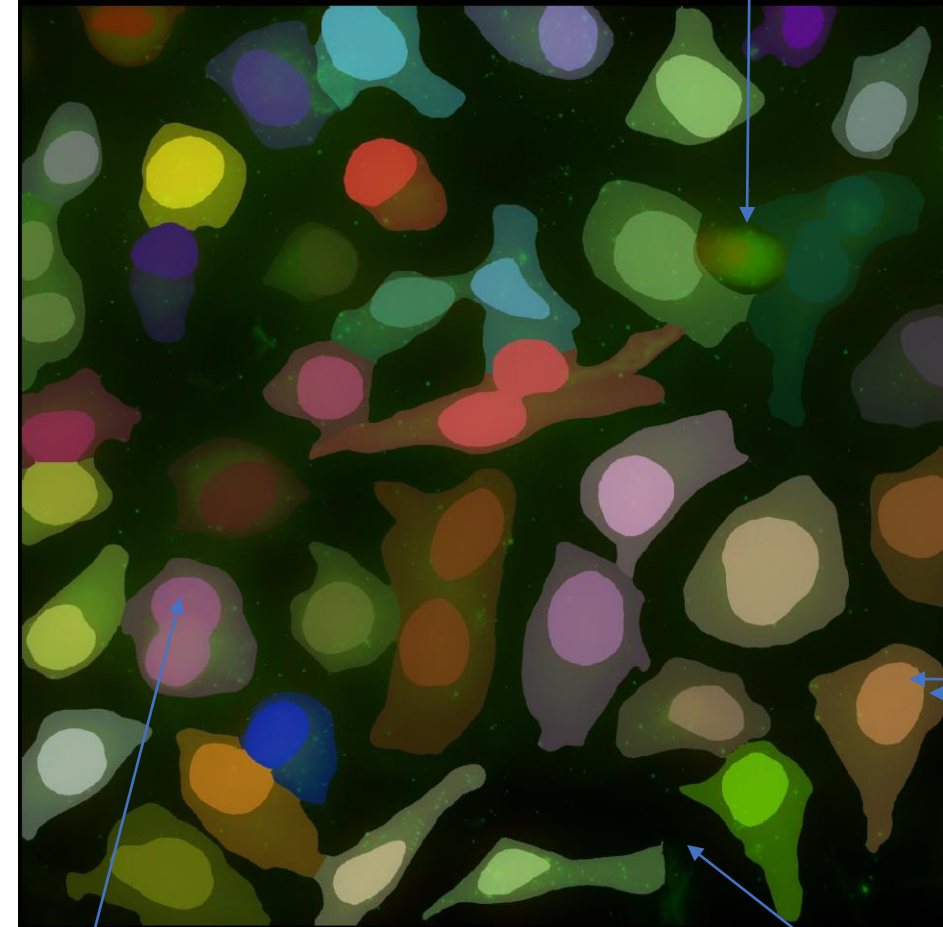
Nuclei without cell

Nuclei without cell: removed



Cell with two nuclei

Cell without nuclei

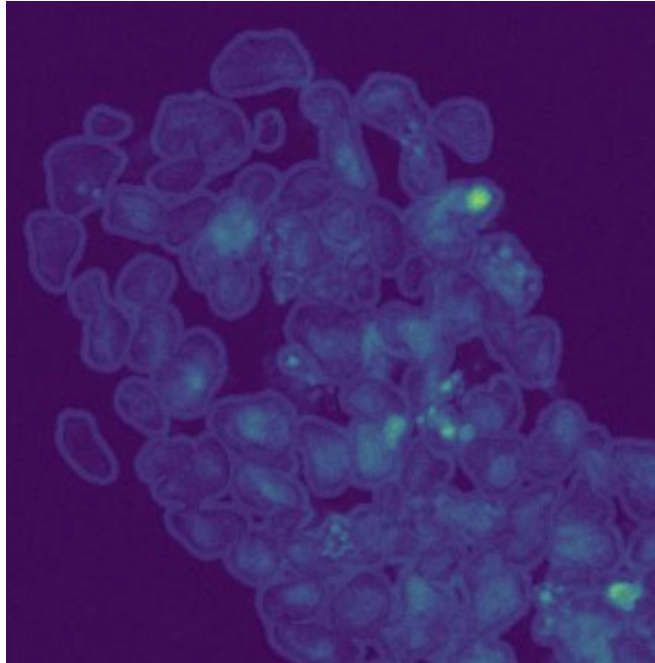


Cell with two nuclei:
Same label

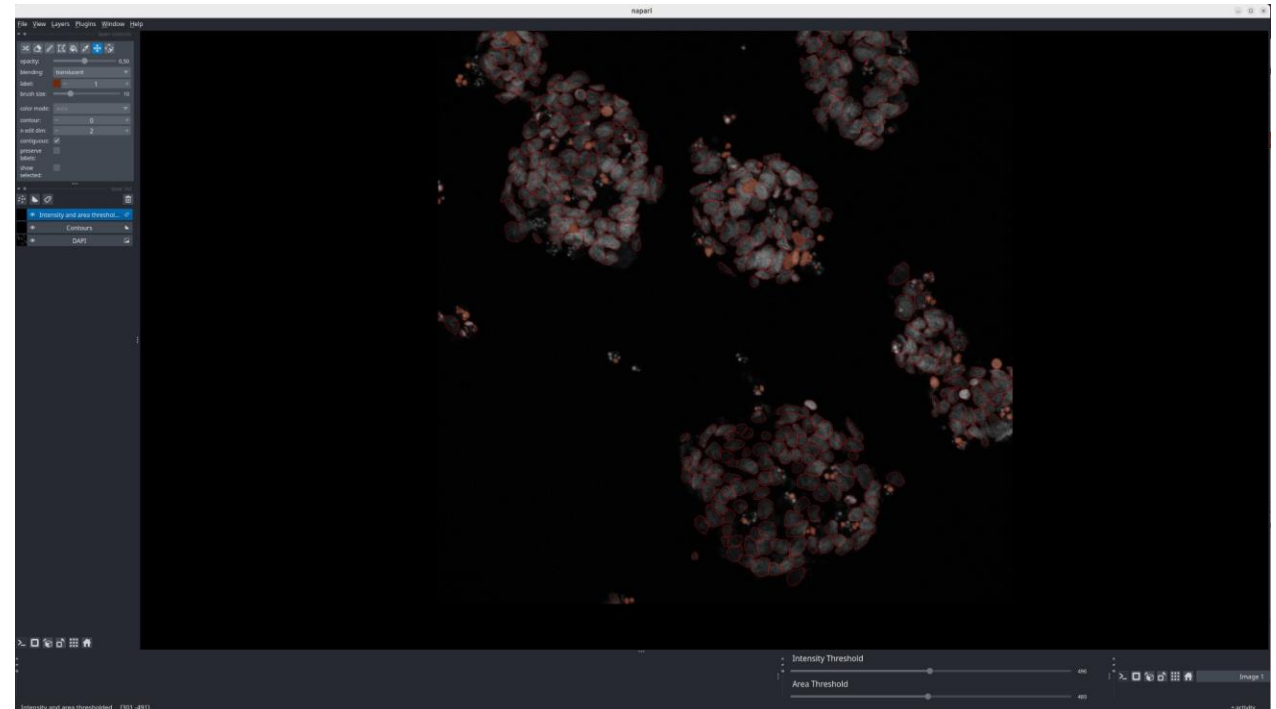
Cell without nuclei: removed

Same index

Segmentation postprocessing (II/II)

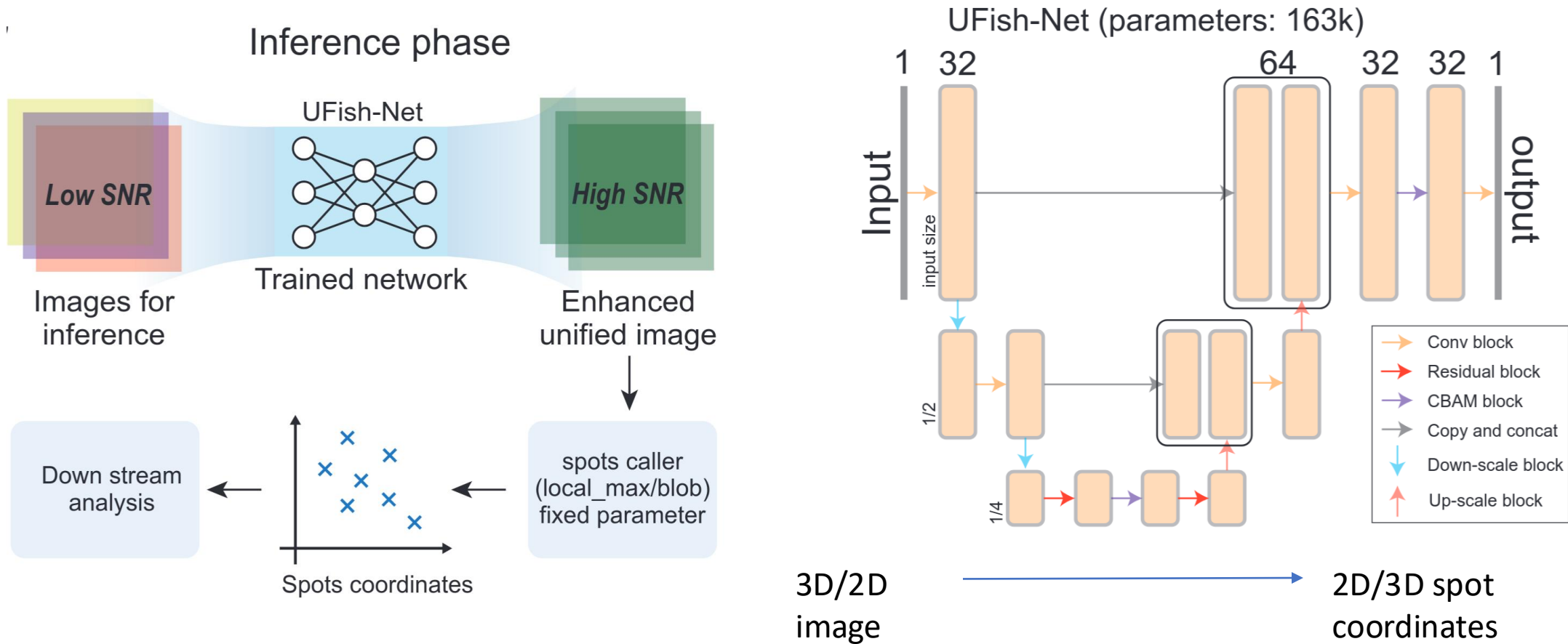


Only the nuclei are visible.
Estimation of the cell body using the
nuclei masks (dilation).



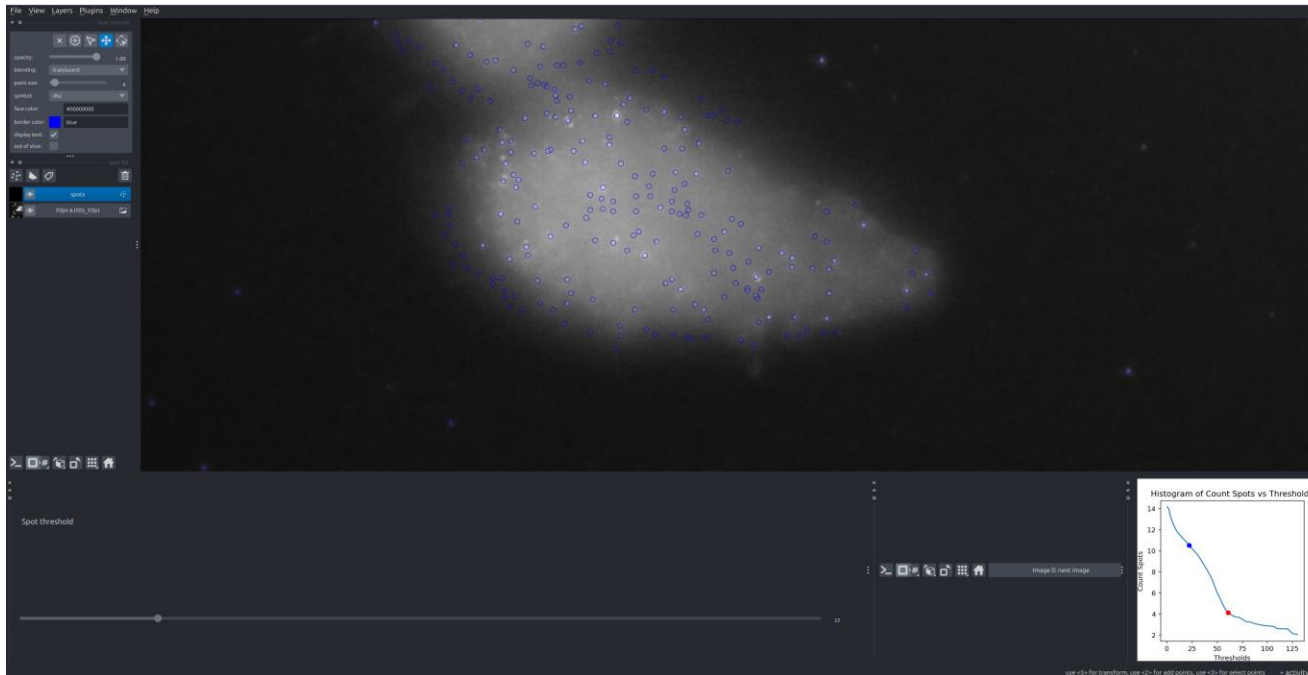
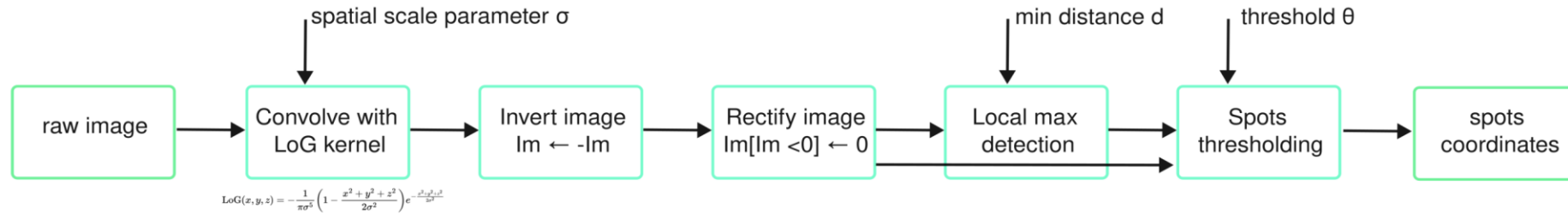
Removing putative nuclei that are too intense
(probably dying cells) or too small.
Napari app (manual displacement of a threshold).

FISH spot detection (I/II): UFISH

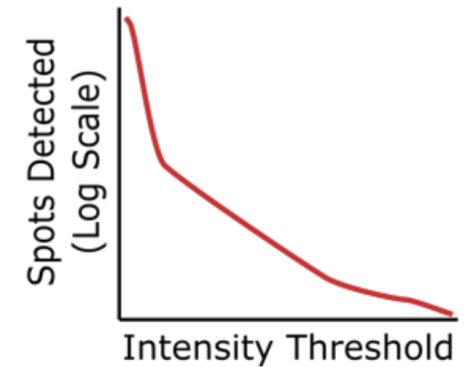


Full automated (no threshold ...)

FISH spot detection (II/II): BIGFISH



*Imbert et al, 2022,
RNA Mueller et al, 2013*

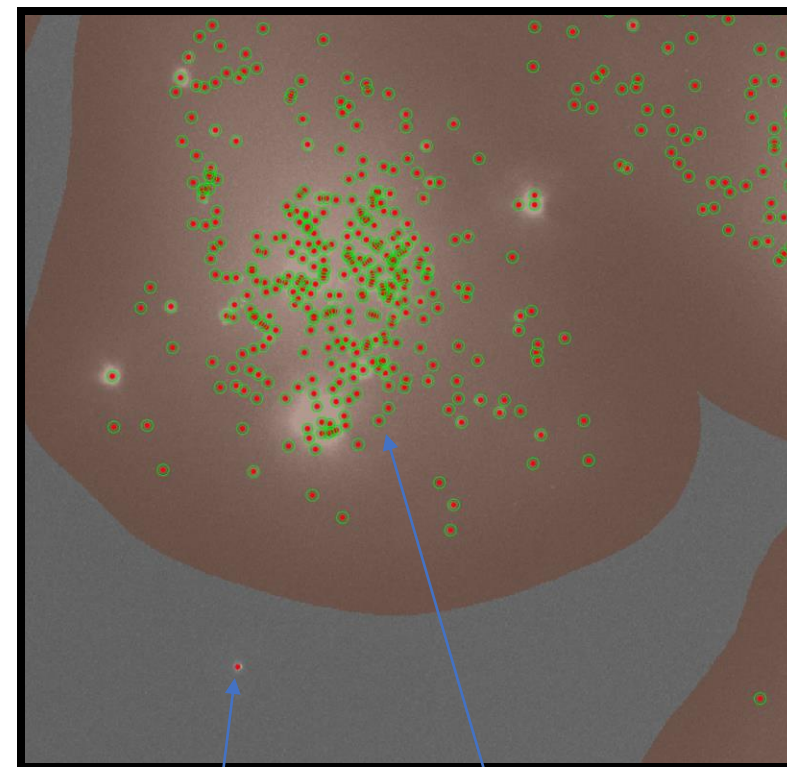
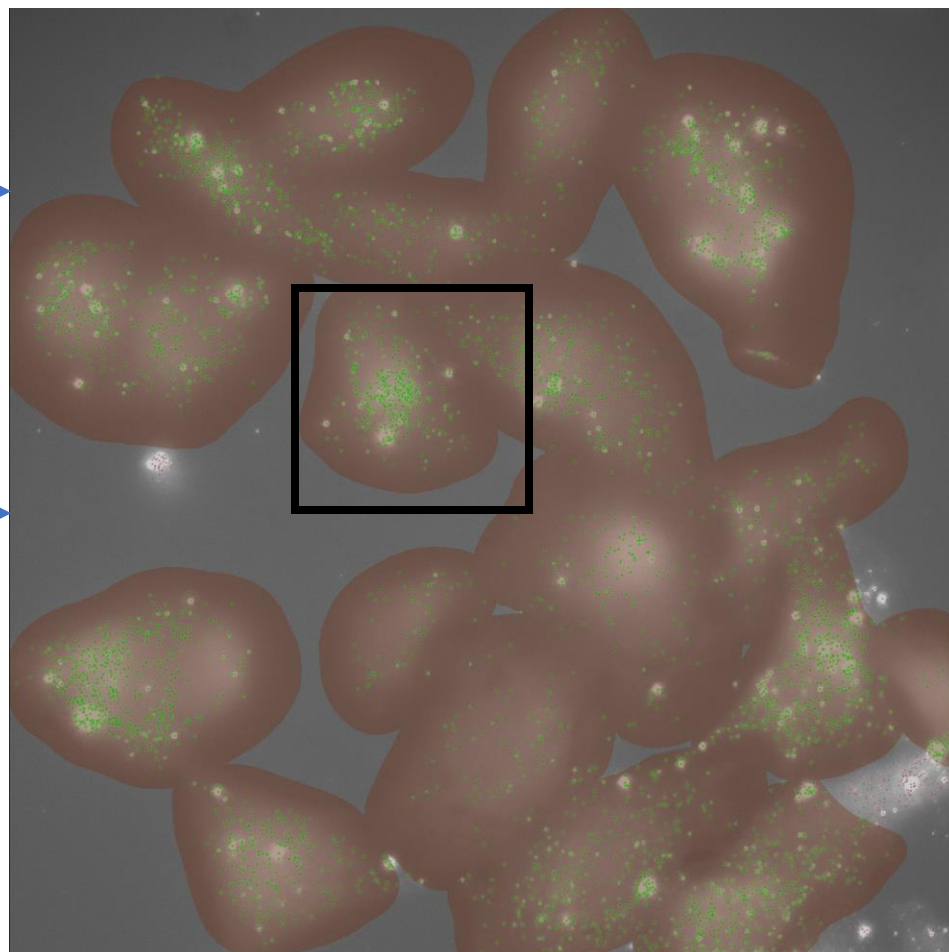


Napari app to threshold semi-automatically using a slider.

Spot cleaning

Cell masks → Binary mask

Spots

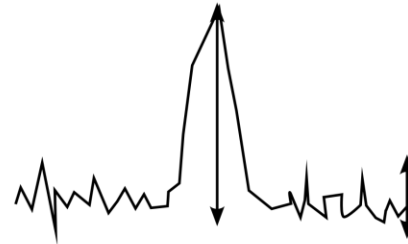
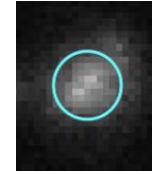


Excluded spot

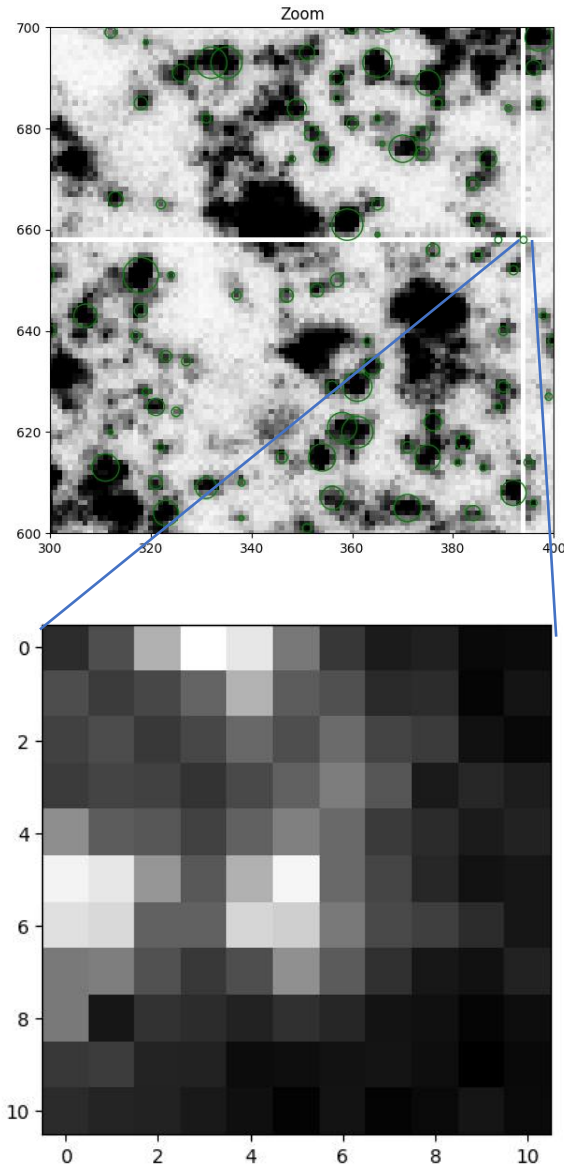
Included spots

RNA quantification: Single RNA descriptors

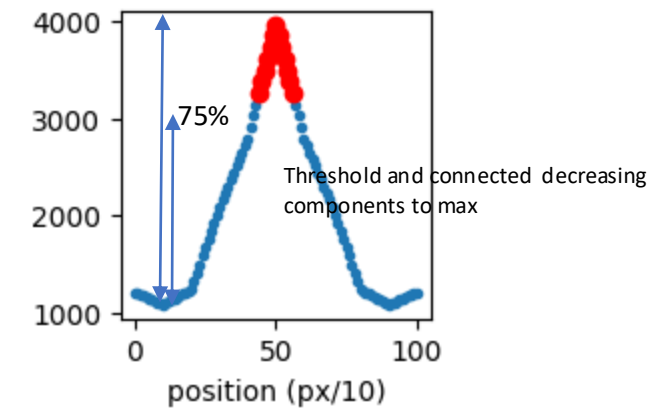
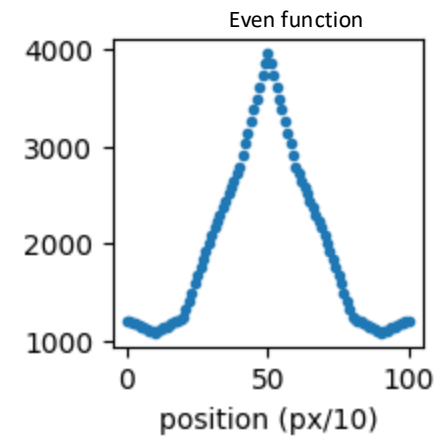
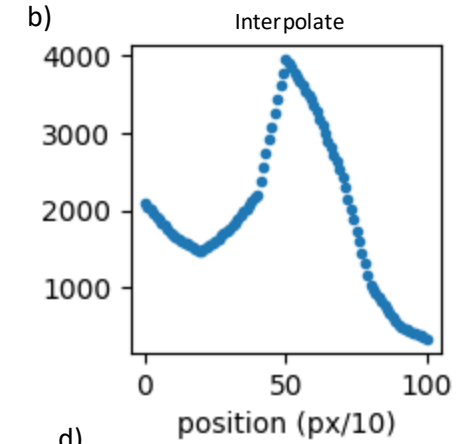
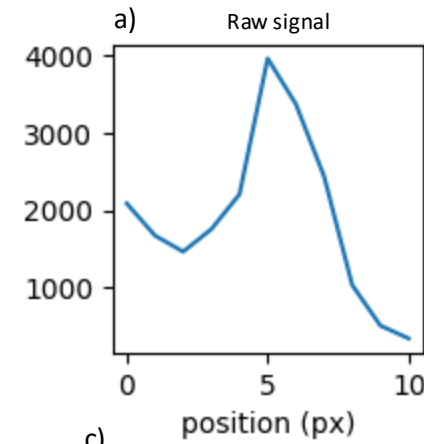
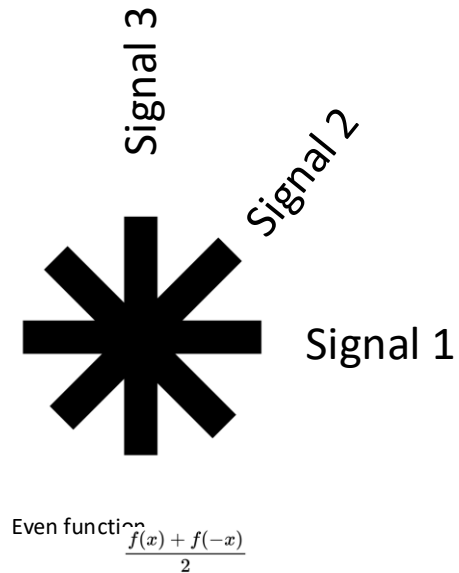
- Spot intensity (take the intensity of the image at the spot coordinate).
- Spot width (in pixels): spot width at the plane of maximal intensity.
- Spot summed intensity : sum all the intensities of the pixels in the circle.
- SNR:
$$\frac{\max(\text{spot}) - \text{mean}(\text{background})}{\text{std}(\text{background})}$$
- Background.
- Cell mask number of the rna.
- In cell (True/False), in cytoplasm (True/False) vs in nuclei (False/True).



Spot size algorithm



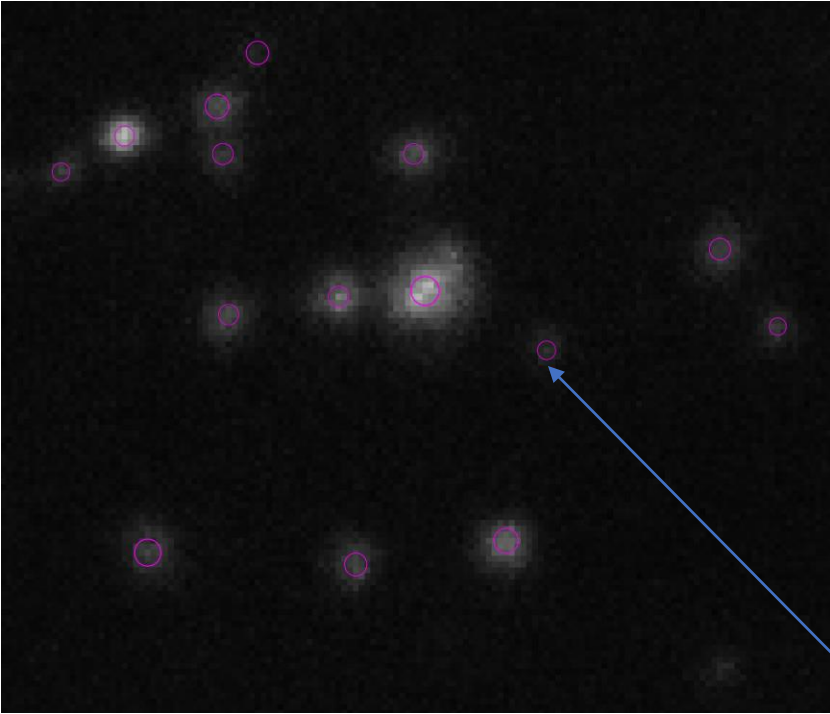
Example for Signal 3:



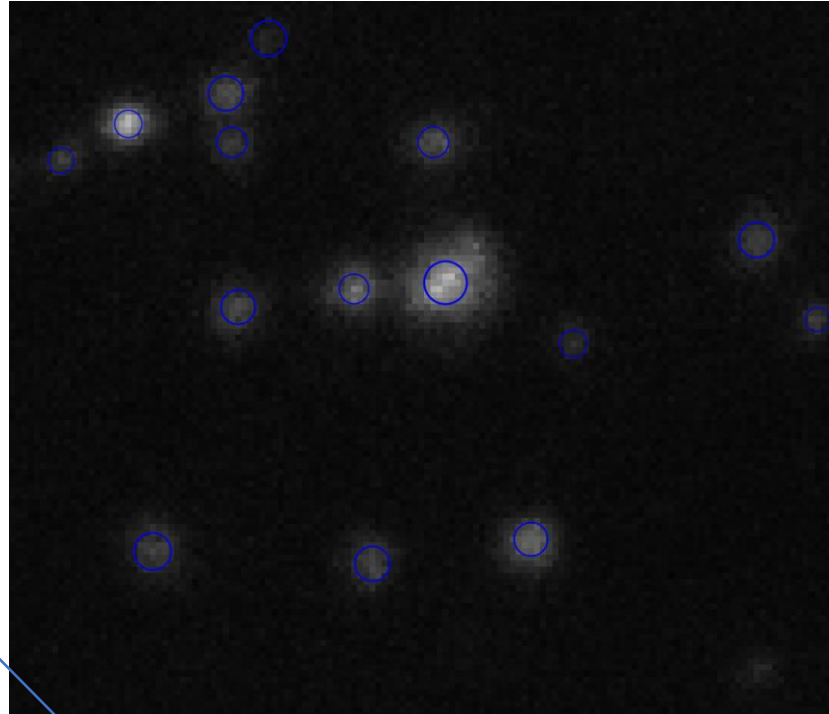
In red: 13 values therefore width3 = 1.3 pixels

width = min(width1, width2, width3)

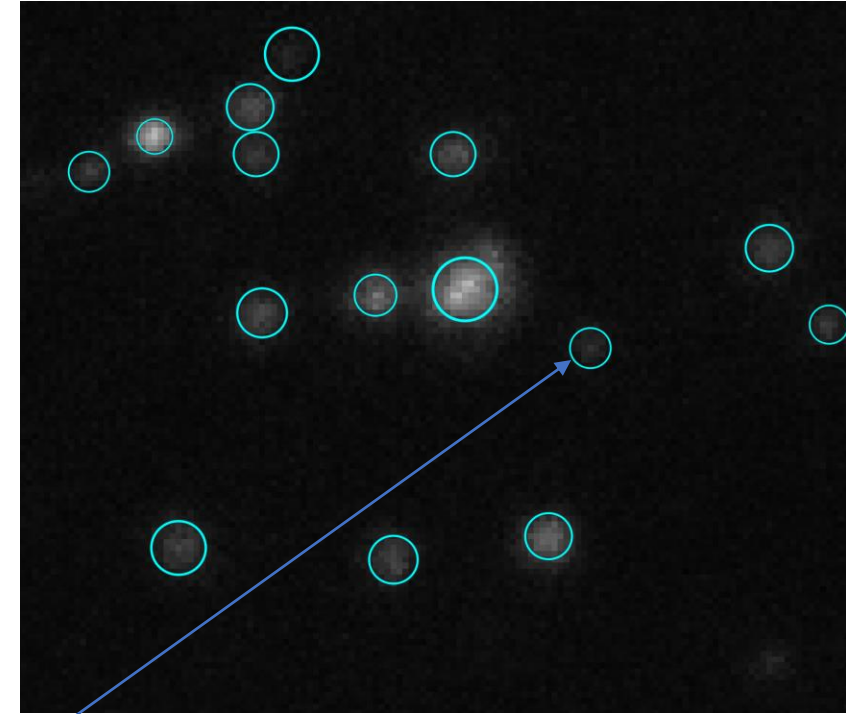
Spots width



Peak fraction = 0.75



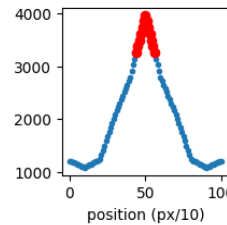
Peak fraction = 0.5



Peak fraction = 0.25

We tweaked the algorithm in such a way that the measure captures better the visual impression.

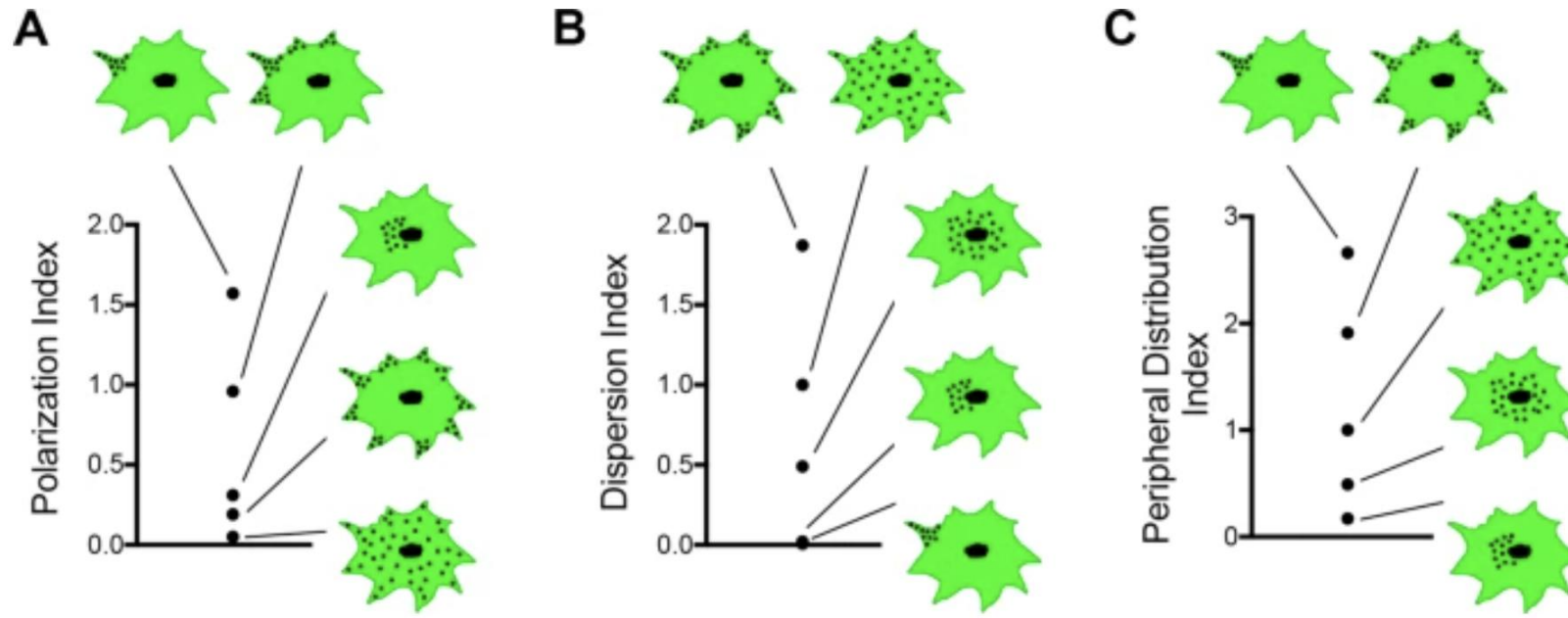
The algorithm of spot width is still imperfect



RNA quantification: Single cell statistics

- For each cell mask:
 - # rnas.
 - # rnas in nuclei.
 - # rnas in cytoplasm.
 - RNA spatial distribution descriptors (polarization index, ...).
 - Colocalization/coexpression (in case in which we have two genes).

Spatial RNA distribution characterization



Stueland, .et al Cells. *Sci Rep* (2019)

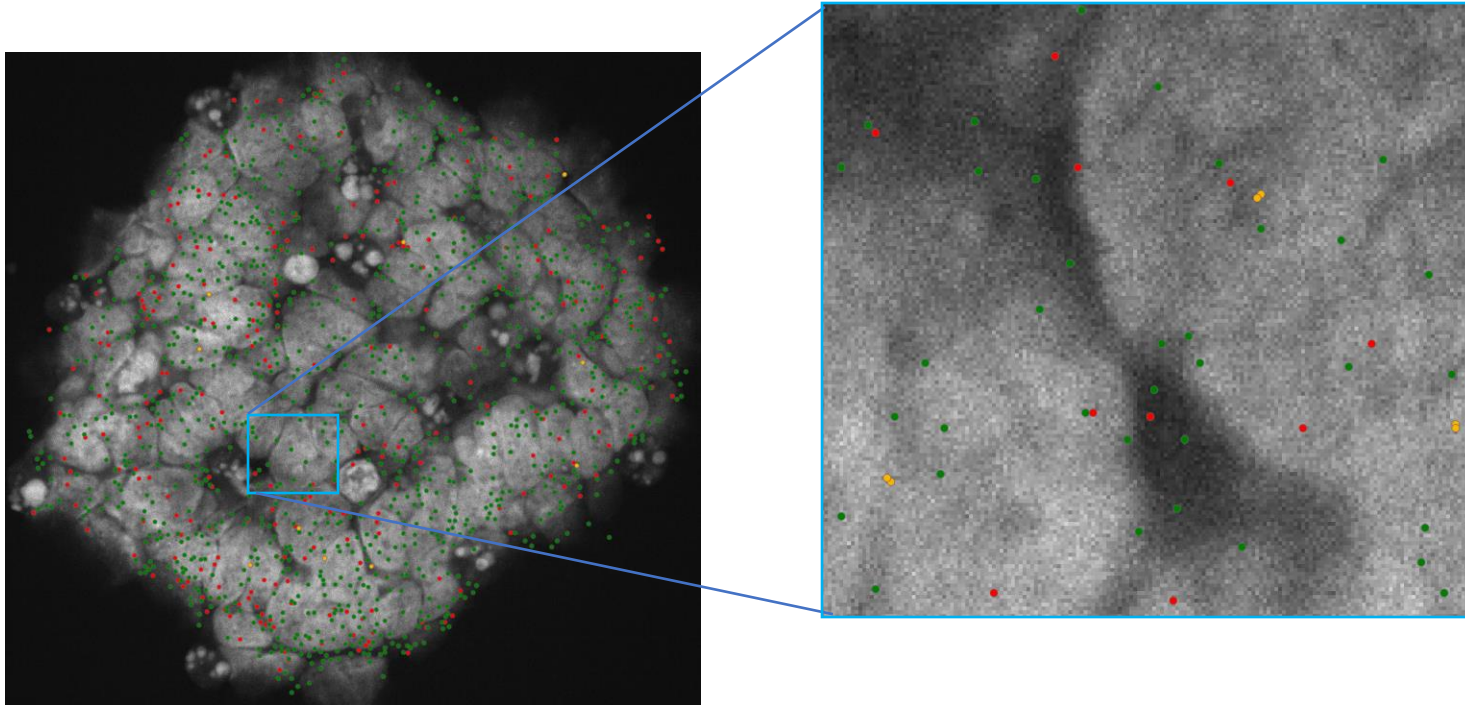
In order to characterize the spatial distribution of rna, we use:

- **The polarization index:** identify the RNA centroid and measure its displacement from the cell centroid (divided by radius of gyration).
- **The dispersion index:** the second moment of RNA pixel intensity positions relative to the centroid of the total RNA signal.
- **The peripheral distribution index:** similar to the dispersion index, but the second moment of RNA pixel intensity positions is calculated relative to the centroid of the nucleus.

$$PI = \frac{\sqrt{(\bar{x}_{RNA} - \bar{x}_{cell})^2 + (\bar{y}_{RNA} - \bar{y}_{cell})^2}}{Rg_{cell}}$$

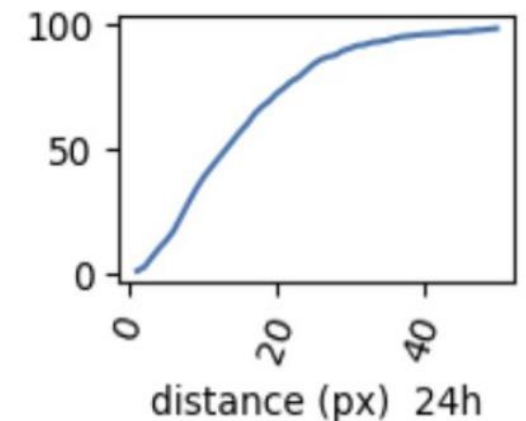
$$\mu_2 = \sum_{i,j} r_{ij}^2 \frac{I_{ij}}{\sum_{i,j} I_{ij}}$$

Coexpression and colocalization of two genes



- Coexpression: for each cell, determine if the two genes are present or not.
- Colocalization: whether, at a given distance, two genes are located.

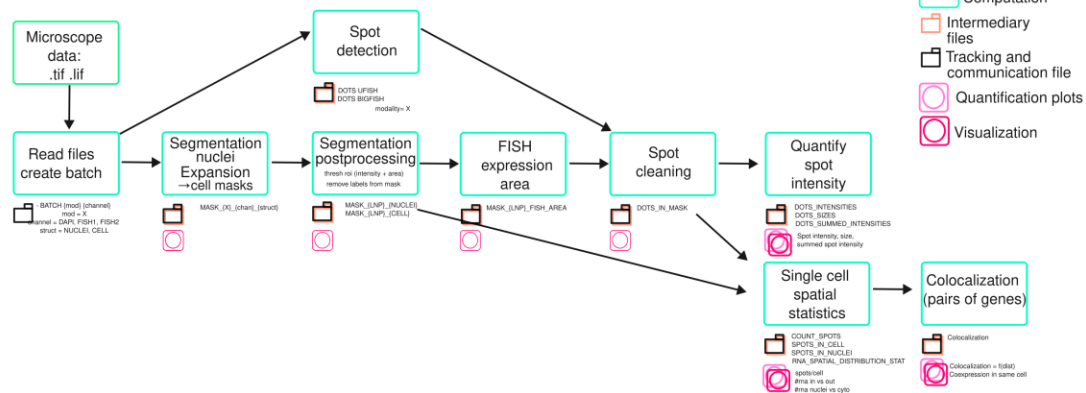
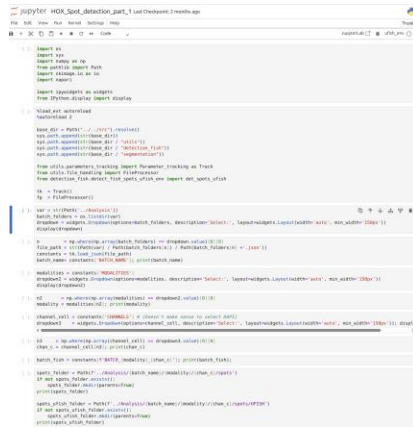
$$colocalization_{G_1} = 100 \frac{card(G_1 \cap G_2)}{card(G_1)}$$



Data pipelines

Nedelec lab: HOX gene expression under different growth factors

Jupyter notebook

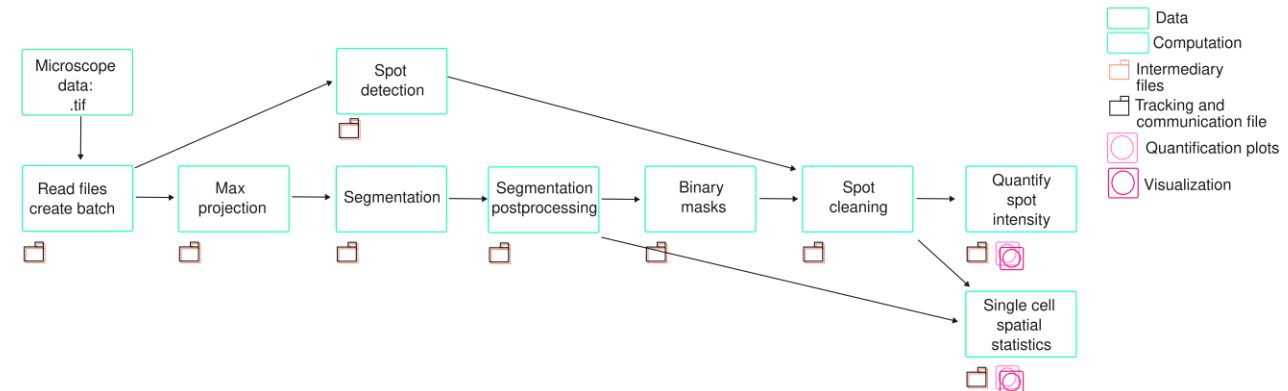


Sanofi: rna detection (LNP)
High content screening



Main loop with handlers

```
0. Graphical overview of the pipeline
1. Define experiment, microscope and data
2. Spot detection
3. Max projection
4. Segmentation
5. Segmentation post-processing
6. Fish area of expression
7. Spot cleaning
8. Quantify spots intensity
9. Single cell spatial statistics
10.Exit application
-----
Enter your choice (0-10): █
```



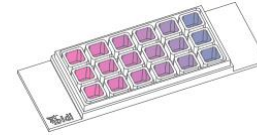
High content pipeline details

- **Code base**
 - 8200 lines of code, 182 functions.
 - Parallelized: scalable, adapted for high content screening.
 - Can work with data on external discs (servers), suited for large data volumes.
- **Most computationally intensive steps**
 - **Segmentation**: Scales with number of fields of view. GPU.
 - **Single spot features** (intensity, snr, background): scales with number of spots. Parallelized.
 - **Single cell spatial statistics**: (# rna/cell, spatial distribution indexes). Scales with number of spots and number of fields of view. Parallelized.
- **Currently supported well plate formats**
 - 18 wells (3 lines x 6 columns) row = A,B,C col = 1,...,6.
 - 96 wells (8 lines x 12 rows). Saved under the convention row = A,B,C..., H, col = 1,...,12.

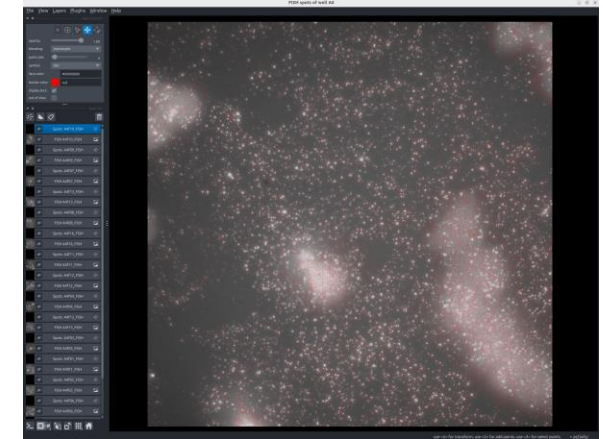


High content pipeline details

- Analysis of 18 wells (16 fields/well, 94 GB) in about 12 hours.



- QC: visual control of each processing step using interactive Napari windows.



- Tracking of the analysis parameters.

JSON	Données brutes	En-têtes
Enregistrer	Copier	Tout réduire
Tout développer		Filtrer le JSON
15:		"A2"
16:		"CS"
17:		"A3"
▼ STRUCTURES:		
0:		"NUCLEI"
1:		"CELL"
▼ METHODS:		
0:		"BIGFISH"
▼ COLORS:		
0:		"blue"
1:		"green"
CHANNELS_NUM:		2
BATCH_NAME:		"test"
▼ CHANNELS:		
0:		"DAPI"
1:		"FISH"

- Output files: 4DN FISH Omics Format, a community data format, based on .csv, that facilitates downstream analysis.

Quantifications: statistics can be computed on demand: single field of view, aggregates of fov, single well, aggregates of wells.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	RNA_Spot_ID	Z	Y	X	in_mask	intensity	spot_width	sum_intensity	snr	background	cell_mask_num	in_cell	in_nuclei	in_cyto
2	1	0	185	1651	False									
3	2	0	196	1624	False									
4	3	0	201	1635	False									
5	4	0	196	1613	False									
6	5	0	199	1594	True	870.0	9.1	57217.0	1.6374901896425744	672.3045808966862	5.0	1.0	0.0	1.0
7	6	0	205	1648	False									
8	7	0	204	1608	False									
9	8	0	208	1600	False									
10	9	0	208	1617	False									
11	10	0	206	1591	True	904.0	8.5	45768.0	1.6572951820634236	698.4751461988304	5.0	1.0	0.0	1.0
12	11	0	208	1571	True	954.0	5.9	30799.0	1.330768859286834	799.3365009746589	5.0	1.0	0.0	1.0
13	12	0	212	1557	True	1023.0	5.5	25043.0	1.1351715127443978	892.7838693957115	5.0	1.0	0.0	1.0
14	13	0	211	1564	True	975.0	4.3000000000000001	15835.0	1.105210008641838	848.4220272904483	5.0	1.0	0.0	1.0
15	14	0	211	1626	False									
16	15	0	212	1586	True	926.0	3.45	11013.0	1.0694705949127656	770.5555555555555	5.0	1.0	0.0	1.0